Viral Infections and Pathogenesis of PCT

There is a well-documented association between porphyria cutanea tarda (PCT) and hepatitis C virus (HCV) infection, as well as combined HCV and HIV infections. The exact role of these viral infections in the pathogenesis of PCT remains unclear. Some studies have suggested a possible interaction with the **HFE gene mutation H63D**, although it is also plausible that the association is secondary to non-specific hepatotoxic effects of the viruses rather than a direct causative role.

Multifactorial Pathogenesis

The known effects of alcohol, estrogens, chlorinated hydrocarbons, iron overload, and viral infections on the heme biosynthetic pathway indicate that each of these factors can contribute to the excessive hepatic porphyrinogenesis observed in PCT. Hepatic siderosis (iron accumulation in the liver) appears to be a critical pathological endpoint in PCT. It is hypothesized that these various factors intensify the ability of iron to inhibit **UROGEN decarboxylase**, particularly by attacking its catalytic site.

Thus, the clinical expression of PCT results from the interplay of multiple genetic and environmental factors. Importantly, unlike in the acute porphyrias, drugs known to induce acute neurovisceral attacks do not typically trigger similar neurological crises in PCT.

Clinical Manifestations

Cutaneous manifestations of PCT are characterized by **vesicles and bullae** that progress to erosions and crusting, primarily in areas prone to trauma—most commonly the **dorsa of the hands** (Fig. 132-8). Skin fragility is a hallmark feature. As lesions heal, scarring may occur, and **milia**—small, pearly white to yellow cysts—frequently develop, especially on the fingers and hands. These skin findings are shared among the hepatic porphyrias with photosensitivity: **PCT, variegate porphyria (VP), and hereditary coproporphyria (HCP)**.

Although lesions predominantly affect sun-exposed areas, patients often do not recognize sunlight as a trigger, since the intense, painful photosensitivity seen in erythropoietic porphyrias is uncommon in PCT. However, many patients report seasonal variation, with worsening in spring and summer and improvement in fall and winter—consistent with increased porphyrin excretion during warmer months.

Additional Dermatological Features

- **Pigmentary changes**: Hyperpigmentation and hypopigmentation may occur, often mottled and resembling **chloasma**.
- **Facial suffusion**: A purplish-red ("**heliotrope**") discoloration may appear on the central face, particularly around the periorbital regions. This finding can mimic the facial plethora seen in **polycythemia rubra vera** (Fig. 132-9) and is not observed in porphyrias of bone marrow origin.
- **Hypertrichosis**: Non-virilizing excessive hair growth, particularly on the face and temples, is a distinctive feature that often prompts female patients to seek dermatological evaluation (Fig. 132-10).